



University of
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GBP/RON Strategy Analysis & Dashboard

Momentum vs Mean Reversion vs Buy & Hold

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Introduction

Do unusually large short-term moves in GBP/RON tend to **continue** (Momentum) or **reverse** (Mean Reversion)? This project tests my personal hypothesis that Mean Reversion should outperform Momentum on 10+ years of daily data, simulating each as a RON-denominated strategy against a buy-and-hold benchmark.

The project also includes an interactive Streamlit dashboard that replays the backtest over the 10+ year period, allowing the user to change the simulation's parameters and analyze how results vary.

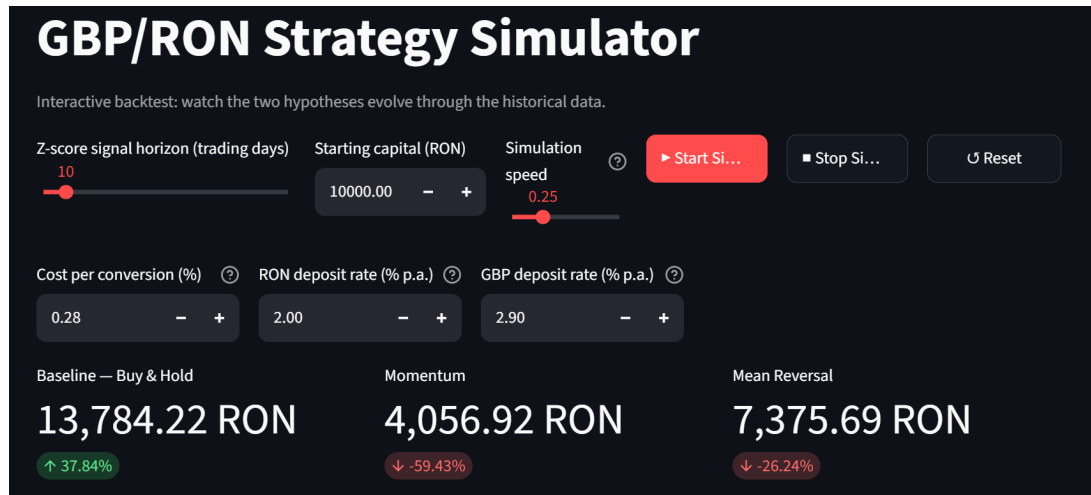


Figure 1: Dashboard Header

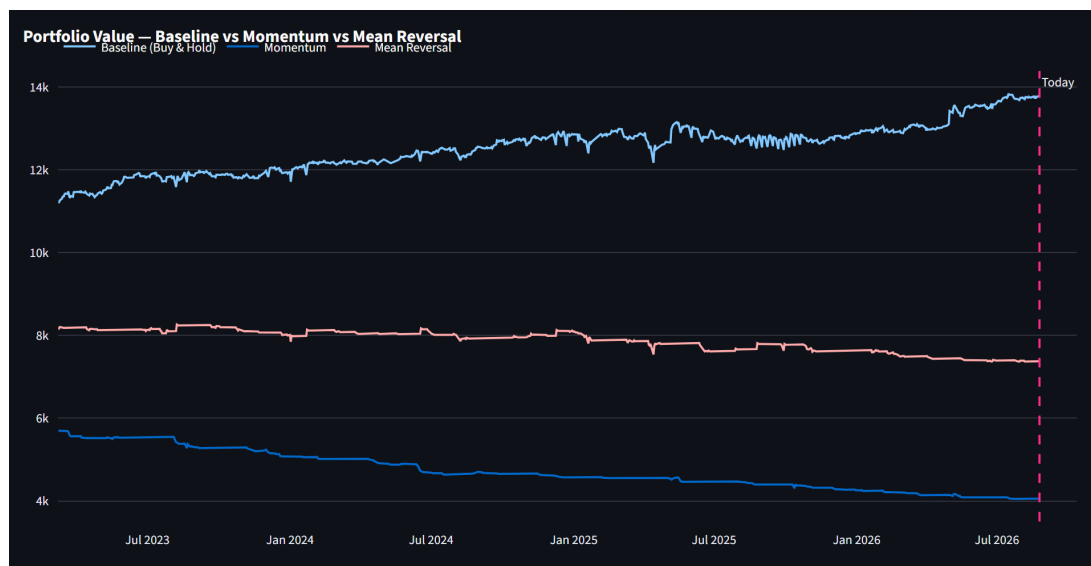


Figure 2: Dashboard Strategy Comparison Graph

AI Usage Note

During this project, I used LLMs (ChatGPT (GPT-5.6 Luna), Claude (Sonnet 5 Extra), Gemini (3.5 Flash)) for:

- *LEARNING*

- researching the appropriate setup for the simulation (how to source data, difference between a short and flat position, difference in volatility and bid-ask spreads between RON and GBP)
- understanding all the factors used to evaluate a strategy aside from the return (Sharpe ratio vs Information ratio, time in market),
- understanding the limitations brought by some of my assumptions (e.g. constant transaction fee vs bid/ask spread, high number of transactions cutting down returns of strategies)
- *CODING*
 - creating the Streamlit dashboard
 - making parts of Python code more efficient and compressed (calculate days elapsed, vectorized calculation of strategy returns)
- *STYLING*
 - comment and output formatting
 - ensuring code is easy to read and consistent across sections
 - formatting README and LaTeX documents.

See Challenges for the obstacles encountered from using AI.

Project Setup

Idea

The inspiration behind this project arose from the necessity to exchange money held in RON, my home currency, into GBP, ever since commencing my studies in the United Kingdom. Thus, on a weekly basis, I analyse the GBP/RON FX rate and decide when and how much of my monthly allowance to exchange.

Navigating this process over periods of political conflict in my home country and economic unrest in the UK raised many questions about how the timing of my FX exchange affects my overall funds, and how I could improve my strategy.

I was thus prompted to create this project to answer questions split into two categories:

1. Does the recent FX trend tell me how this will evolve over the next few days, and can this indicate if I should exchange now or wait?
2. What would happen if I could simulate how my funds evolve over a long period of time by using different strategies? What if I assumed I also exchange GBP back into RON - can that help my funds long-term? Do some strategies protect my funds over periods of global uncertainty and conflict, and how much can I generalise?

This project aims to bring together the main factors involved in currency exchange and evaluate how different strategies perform over time. Key aspects when choosing to exchange or hold include the **FX rate**, the **interest** earned on funds in each currency and the **transaction costs**.

Another reason for creating the project was my desire to improve my understanding of how to quantify all the variables that go into trading currencies, and how to model the relationships between them.

Data

- **Source:** Yahoo Finance via `yfinance`, ticker GBPRON=X
- **Frequency:** used daily *closing* prices
- **Sample:** January 2016 to September 2026 (10+ years)

Signal

As previously mentioned, the rationale behind choosing to exchange or not lies in the recent trend in the GBP/RON exchange rate. As such, the **signal** used to decide if and when to exchange is the *average return of the FX rate* in the past 10 days, standardized against its recent 40-day behaviour. I chose this

signal to capture the recent FX trend over 2 trading weeks, which I considered enough time to capture a real trend, not merely daily volatility.

The rationale for standardizing it is to compare a recent trend with the longer-term, yet still recent market behaviour - so as to compare the recent trend against the current climate, not the entire dataset. This is rooted in the assumption that the RON will naturally lose purchasing power over time compared to GBP, because of Romania's highly elevated inflation and weaker economy compared to the United Kingdom's.

Formulas used for the signal:

```
r_t = Close_t / Close_(t-10) - 1 # 10-day return
z_t = (r_t - mean(r over 40 days)) / std(r over 40 days) # standardized
```

Note: The 10- and 40-day windows *overlap*, meaning that consecutive observations are *not independent*, which can affect the statistical conclusions drawn from the project. The motivation behind the *10-day window* for the return lies in attempting to use a significant trend as a signal, one that persists over a longer period of time - not relying on day-to-day volatility. Additionally, although using *daily returns* would mean having independent data points, using them as a signal would also create very frequent trading opportunities, increasing transaction costs and consequently reducing the strategies' returns. Given more time, I would research a way to use independent windows of time without capturing daily volatility and increasing transaction costs.

Hypotheses & Strategies

The strategies are defined as such:

	Idea	Rule
H1 - Momentum	An unusually strong trend in GBP/RON continues	Hold GBP while $z > 1$, otherwise hold RON
H2 - Mean reversion	An unusually strong trend in GBP/RON reverses	Hold GBP while $z < -1$, otherwise hold RON

Both strategies use the same z-score calculation of the recent GBP/RON return standardized against its own recent history. A standard deviation of 1 is used as the threshold, to allow for trends to be identified and used as a signal even if they are not very extreme.

The **benchmark** is **buy-and-hold**: convert all RON to GBP on day one and never trade, allowing the GBP to accumulate interest.

The strategies are tested **separately**, never combined.

Each strategy either holds GBP or sits in RON; neither goes short.

Part 1 - Does the signal have predictive power?

For every day, the forward 10-day return (close t to close $t+10$) is averaged across the three categories as a simplified measure of short-term strategy performance:

- Momentum: $z > 1$,
- Mean Reversion: $z < -1$,
- Baseline: $-1 \leq z \leq 1$.

This provides an estimate of the expected return over two trading weeks if today's signal were acted upon.

Note: The baseline here differs from the baseline in Part 2 - in Part 1, we need a benchmark to compare the future average returns with, which is the in-between of the two strategies (average return if acting on neither extreme). In Part 2, we need a benchmark strategy for the simulation, also to compare the results of the "active" strategies with - but it does not map 1-1 with the benchmark from Part 1.

Part 2 - Portfolio simulation

- Each strategy starts with **10,000 RON**.
- **Timing:** the z-score at the close of day $t-1$ sets the position held on day t , which earns the return from close $t-1$ to close t .
- **Daily strategy return** = $(1 + \text{GBP/RON return}) * (1 + \text{GBP_Interest}) - 1$ if position holds GBP (position exposed to FX fluctuations and accumulates interest) ; $(1 + \text{RON_Interest})$ if position holds RON (only accumulates interest); exchange fees also applied on days when trade occurs; portfolio value compounds these returns.
- **Metrics reported:** *final value*, *total return*, annualised *Sharpe* ratio and *Information* ratio (both compared to Buy&Hold benchmark, see Notes for explanation of benchmark), number of *days holding GBP*, and number of *entries and exits*.

Note: Exchange Fees & Interest Rates

To better simulate a real-world scenario, the project implements transaction costs on each entry/exit and interest rates while holding each currency:

- Transaction Cost = 0.0028
- RON Interest = 0.02 p.a.
- GBP Interest = 0.029 p.a.

The sources for the values are:

- Revolut's exchange fee (weighted average between 0% on weekdays, 1% on weekends; $\frac{2}{7} \approx 0.286\%$)[1],
- Revolut's interest rate on GBP instant access savings account[2],
- BCR's (Romanian Commercial Bank) interest rate on instant access savings account[3].

Since these are the banks I use to exchange and hold my funds, I found it appropriate to use them as sources for these values in the simulation.

Notably, this implies using retail deposit interest, which is not realistic for an FX trading portfolio, but closely mirrors my real-life situation - see Limitations for how these assumptions are simplifications of a real-world scenario.

Dashboard

An interactive Streamlit app that replays the backtest through history one trading day at a time.

Note: Repeated runs of the project may cause Yahoo Finance to throw an exception when attempting to retrieve the data. In this case, the DataFrame may be empty and the dashboard will be unable to load. If this occurs, wait before trying again and clear the cache or restart Streamlit.

Features:

- Adjustable z-score lookback (5–60 trading days), starting capital and animation speed
- - Adjustable Exchange fee, GBP Interest, RON Interest
- Start / Stop / Reset controls
- Live portfolio values and % returns for Buy & Hold, Momentum and Mean Reversion
- Single Plotly chart plotting the three strategies with a marker for the simulated “today”

Results

Part 1 Results - Analysis

Reminder: Part 1 only analyses future 10-day returns for each strategy. This does not model transaction fees or interest in either currency - the simulation in Part 2 includes these.

Strategy	Signal	Average forward 10-day return
Momentum	$z > 1$	-0.0966665
Mean reversion	$z < -1$	0.268212
Baseline	$-1 \leq z \leq 1$	0.0017033

Observations

We notice the Momentum strategy had the lowest average future returns, followed by the baseline, with Mean reversion giving the highest returns. This first insight provides some basis to support the initial hypothesis that exchanging during a decreasing trend will be more profitable in the short-term future.

Notably, Part 1 suggests that Mean reversion outperforms not only the momentum strategy, but the baseline too - which we may keep in mind going forward, to compare with Part 2's results.

However, we remember that Part 1 is the theoretical, simplified framework used to initially define and compare my hypotheses - it does not include transaction costs or interest rates, so we refer to Part 2 for a more comprehensive analysis.

Part 2 Results - Portfolio Simulation

Default Parameters (can be modified via Streamlit dashboard):

- Initial Capital = 10,000 RON
- Z-score lookback period = 10 days - Transaction Fee = 0.28%
- RON interest = 2% p.a.
- GBP interest = 2.9% p.a. *

Note: See Limitations for how the assumption of constant fees and interest rates over time can affect the simulation's viability.

Results with default parameters:

Strategy	Final value (RON)	Total return	Sharpe	Information	Time in GBP	Entries / exits
H1 Momentum	4,061.77	-59.38%	-2.78	-1.32	17.51%	344
H2 Mean reversion	7,384.50	-26.16%	-1.22	-0.73	18.84%	300
Buy & hold	13,810.18	38.10%	-	-	100%	1

Observations

1. The Momentum and Mean Reversion strategies both underperformed the **Buy & Hold** benchmark, noting their significantly negative total returns.

2. Additionally, comparing final **Momentum** and **Mean Reversion** values, we observe:

- H1's return is negative and over twice as large as H2's,
- H1's Sharpe is also negative, with an absolute value more than twice that of H2's,

- H1 has less time in market but more entries/exits, suggesting short periods of holding GBP and frequent currency exchanges, which consequently increases transaction costs,

We may thus conclude the Momentum strategy underperformed Mean reversion over the entire dataset. This conclusion provides numerical evidence to support the hypothesis I started the project with. However, it is interesting to see this conclusion does **not** hold over all periods of the data (see bullet point 4).

3. The drastic overall decrease in value of the Momentum and Mean Reversion portfolios can be attributed to the **large number of currency exchanges** - 344 and 300, respectively. This indicates the need for a *minimum holding period*, or a *different exit threshold* rather than simply exiting when the entry condition is no longer satisfied, in order to minimize the number of trades and consequently minimize the losses due to exchange fees.
4. The overall results do **not** hold over the entire dataset. The figures below demonstrate how different market periods favour different strategies, which motivates further analysis of whether particular market regimes favour specific strategies:

Figure 3 illustrates how over the first ~18 months of the simulation, the benchmark had the lowest performance of the three, drastically contrasting the final results:



Figure 3: Dashboard: Strategy Results 2016 - 2017

Light Blue - Baseline ; Pink - Mean Reversion ; Dark Blue - Momentum

Figure 4 depicts the *reversal* of this trend around the start of 2018, when both active strategies start underperforming compared to the benchmark:



Figure 4: Dashboard: Strategy Results 2016 - 2019

Light Blue - Baseline ; Pink - Mean Reversion ; Dark Blue - Momentum

Figure 5 then reinforces this trend - around the onset of the COVID-19 pandemic we notice the performance gap widening between the strategies in favour of buy & hold. Notably, all strategies experience higher levels of volatility during the global phenomenon:



Figure 5: Dashboard: Strategy Results 2017 - 2020

Light Blue - Baseline ; Pink - Mean Reversion ; Dark Blue - Momentum

Lastly, Figure 6 captures the continuation of the trend over the following 3 years, with the benchmark experiencing a steady increase, Mean Reversion remaining relatively stable, and Momentum linearly decreasing:

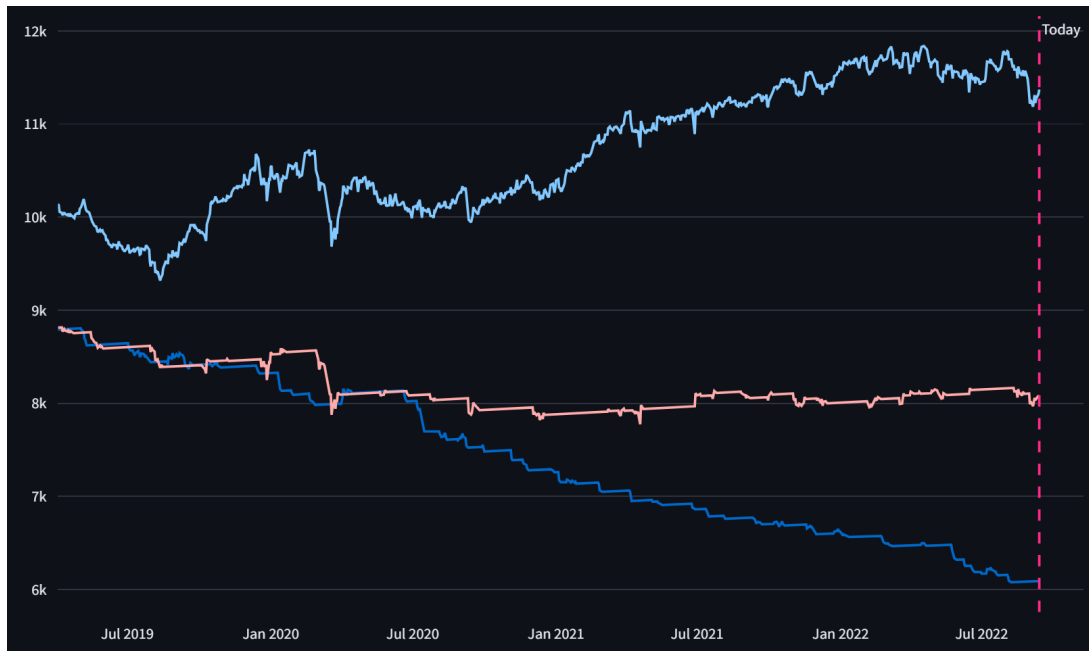


Figure 6: Dashboard 2019 - 2022

Light Blue - Baseline ; Pink - Mean Reversion ; Dark Blue - Momentum

We may thus conclude that there may exist a *relationship between different market conditions and each strategy*, despite the overall results from the 10+ year period indicating that Buy & Hold is more profitable overall.

This also indicates that *conclusions drawn from over 10 years of data may not be clear and may not generalise to other periods*, as such an extended period of time captures numerous political shifts, world conflicts and different socio-economic conditions which favour different attitudes towards risk.

Results from Dashboard with modified parameters

The dashboard allows the following parameters to be modified:

- *Z-score lookback period*: # of days used to calculate the strategies' signal; default = 10; range: 5 - 60
- *Starting capital*: initial sum invested in each strategy, measured in RON; default: 10000
- *Simulation speed*: # of seconds between animation frames; higher = slower animation ; lower = faster animation; default = 0.25 ; range: 0.05 - 0.75
- *Exchange fee*: % charged on every exchange; default: 0.28%; minimum: 0.03%
- *RON Interest*: % p.a. the RON deposit accumulates; default: 2%
- *GBP Interest*: % p.a. the GBP deposit accumulates; default: 2.9%

Changing the simulation parameters via the dashboard yields different conclusions:

1. **Lowering Exchange Fee** (0.03% instead of 0.28%) -> the **Mean Reversion strategy outperforms the benchmark**, but Momentum still doesn't - indicating that reducing transaction costs favours the active strategies, but Momentum still enters and exits the GBP position too frequently:

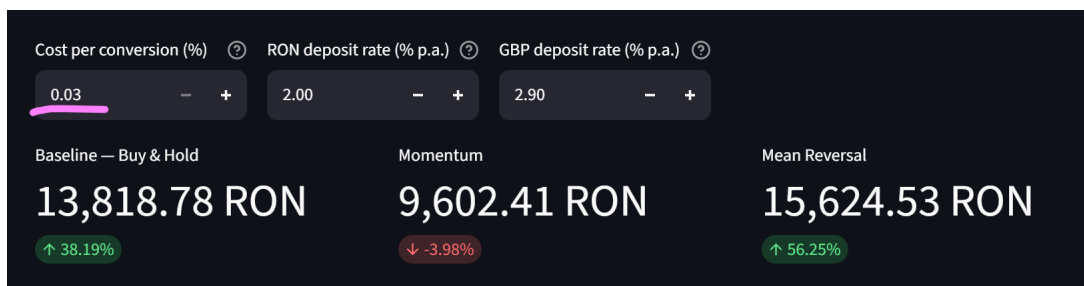


Figure 7: Strategy Results for 0.03% Exchange Fee

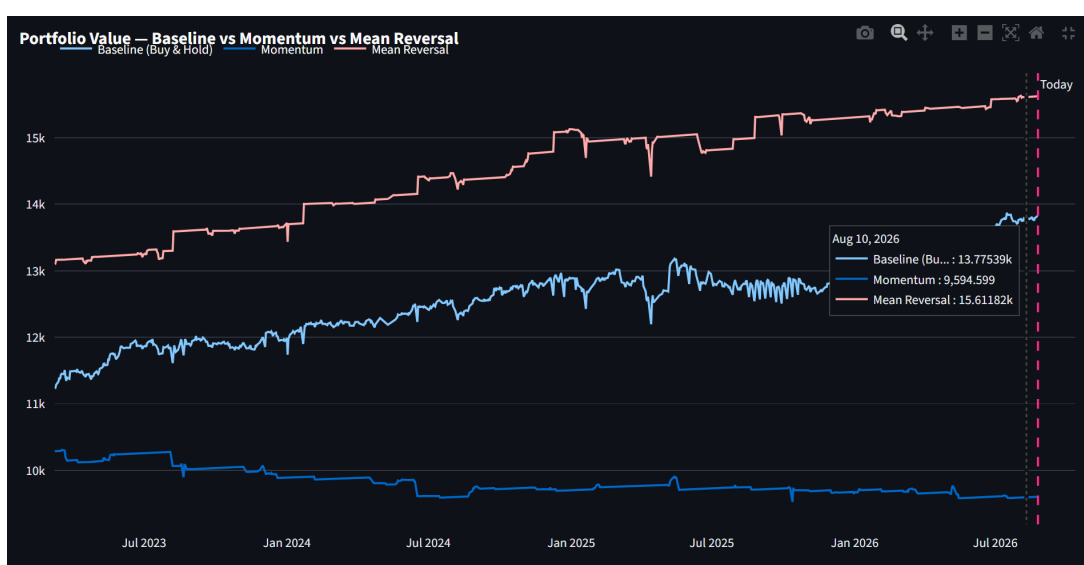


Figure 8: Strategy Results for 0.03% Exchange Fee - Graph comparison

2. **Increasing RON interest rate** (3% instead of 2%) -> increases returns of strategies but not Buy & Hold:

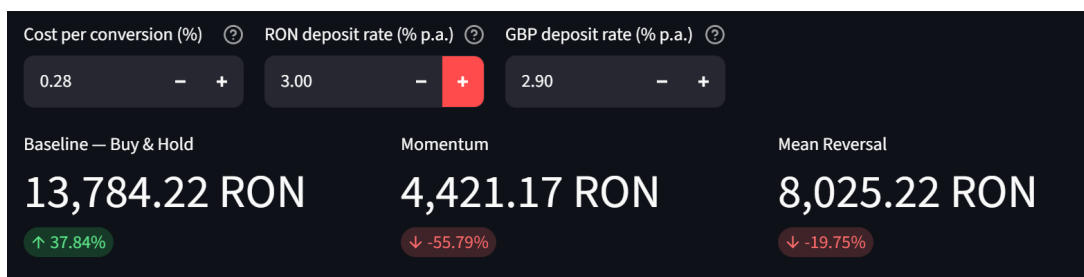


Figure 9: Strategy Results for 3% RON Interest rate

3. **Increasing GBP interest rate** (3.9% instead of 2.9%) -> increases returns of all strategies (but not equally):

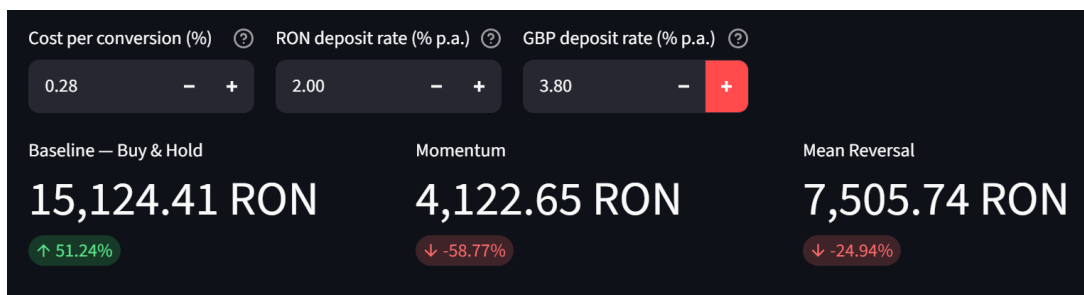


Figure 10: Strategy Results for 3.9% GBP Interest rate

4. Changing z-score lookback period -> The relationship between the z-score lookback period and overall performance is unclear. Changing it seems to affect the strategies differently - some higher values positively impact both active strategies (23 days -> see Figure 9); other higher values impact Mean Reversion positively and Momentum negatively (51 days -> see Figure 10).

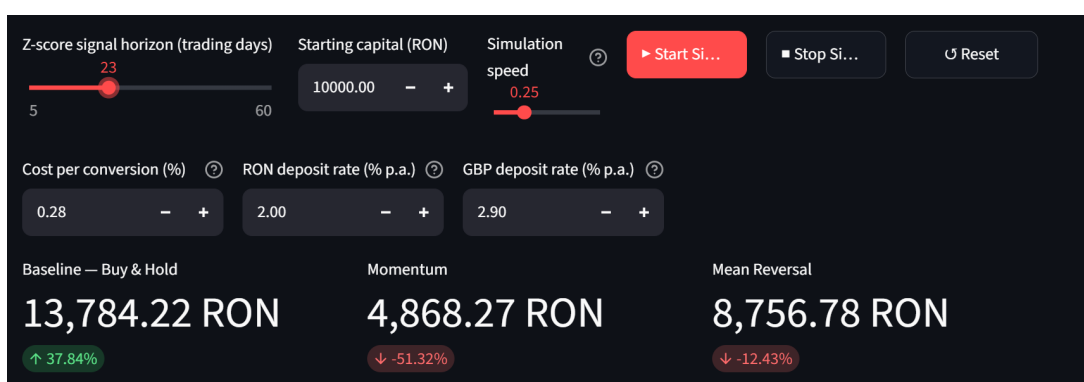


Figure 11: Strategy Results for z-score lookback period of 23 days

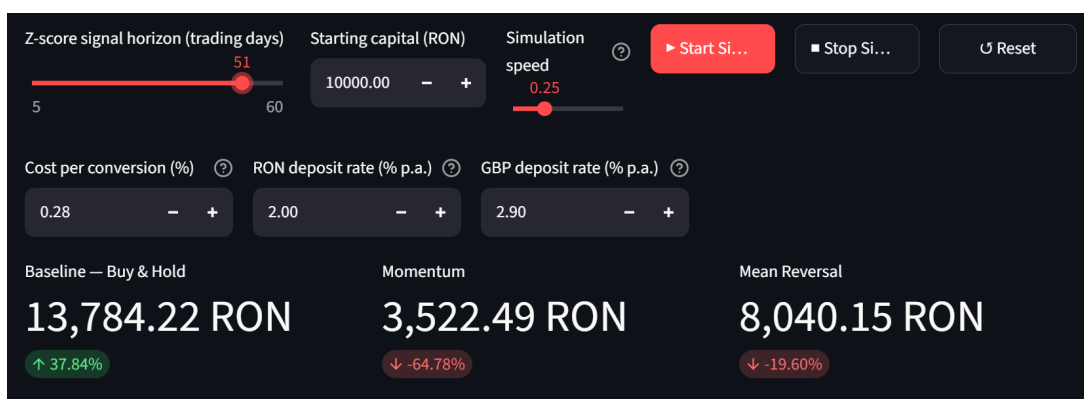


Figure 12: Strategy Results for z-score lookback period of 51 days

The unknown relationship between these parameters and the strategies leaves room for further analysis in future iterations of the project.

Conclusion

Overall, I set out to test my hypothesis that the Mean Reversion strategy should outperform both Momentum and the Buy & Hold benchmark. Although the simplified, theoretical Part 1 provided some numerical results to support this hypothesis, the more complex simulation in Part 2 provided an opposing perspective:

Strategy	Final value (RON)	Total return
H1 Momentum	4,061.77	-59.38%
H2 Mean reversion	7,384.50	-26.16%
Buy & hold	13,810.18	38.10%

Notably, **both strategies significantly underperformed the Buy & Hold benchmark**, which contrasts my initial hypothesis that attempting to use recent market trends can help with the value of my starting funds. However, as noted throughout the project limitations, this numerical result is largely due to the *high number of GBP position entries and exits*, increasing transaction fees and consequently creating such negative returns.

Nonetheless, this project constitutes a great learning opportunity and has improved my understanding of the various factors that drive strategies and taught me how to *define and quantify my hypotheses, understand the data* needed to test them, and *provide an analysis* of the results.

I am keen to gain more knowledge in the field in order to be better equipped to interpret the relationships between the factors and patterns I believe to be relevant and the results I find.

Notes

- **Benchmark for Sharpe and Information Ratios:** Initially, as a benchmark for both the Sharpe and Information ratio, I used the returns of holding RON and allowing it to gain interest - this felt like the natural risk-free alternative since holding RON is not exposed to GBP/RON fluctuations. However, I then realised that the end goal of my portfolio is to hold GBP (as this is supposed to model my real-life scenario where I need GBP to spend in the UK) - so even holding RON has risk, because its value could depreciate over time and have less purchasing power in GBP, which is ultimately what I am interested in. I therefore use the *Buy & Hold* strategy as the benchmark. It represents the passive alternative most closely aligned with the portfolio's objective of ultimately holding GBP. However, it is not risk-free, as GBP remains exposed to exchange-rate and other market risks.
- **Weekend interest rate mismatch:** Over the weekend, there is one FX move (Fri-Mon), and the interest rate accumulates every day over the three days $(1 + \text{daily_rate})^{(3/365)} - 1$ - so, in the data frame, the Monday entry after a weekend contains these two values. Thus, the calculation for applying both interest and the FX move to the GBP portfolio over the weekend assumes the compounded interest rate was applied on the FX change reported on Monday - i.e. it assumes Saturday and Sunday gain interest on Monday's FX change, instead of Friday's. More realistically, all of Friday, Saturday and Sunday have the same FX rate as Friday, so the interest should be applied to this one rate until Sunday, and then once more to the Monday rate. This has been simplified due to limited time.

Limitations

- **Execution timing:** Trades are assumed to execute at the *same closing price* used to compute the signal. Although the signal on day t determines the position on day $t + 1$ (to avoid look-ahead bias), the project uses the ratio $\frac{\text{closing}_{(t+1)}}{\text{closing}_t} - 1$ to calculate the return of each day when holding the position. This means that on the first day of holding the position, the closing price on day t is

used instead of the opening price on day $t + 1$ - which realistically cannot happen because the signal on day t is calculated *after* the market closing on day t , so the exchange could not be executed. However, this is a simplification chosen due to limited time.

- **Identical entry/exit threshold:** The results highlight the key issue with both the Momentum and Mean Reversion strategies - the high number of trades driving transaction fees up. A main effect of this is the signal indicating the strategy should enter/exit the position too often, because of the binary nature of the strategies' designs - for Momentum: hold GBP when $Z > 1$, RON otherwise; for Mean Reversion: hold GBP when $Z < -1$, RON otherwise. Thus, the strategies switch often, increasing the number of trades. This could be improved by having *separate thresholds for entering/exiting the positions*. Take Momentum as an example: enter GBP position when $Z > 1$, exit when $Z < 0.5$ - thus implying the need for a *weaker signal to exit* the position once it has been entered. This leaves room for further analysis around the value of each threshold and what it means for the signal strength once a second threshold is introduced.
- **Constant transaction costs:** Used a constant 0.28% transaction fee per exchange instead of modelling the bid/ask spread. RON is an emerging-market currency that typically trades at wider spreads than GBP - thus, since the strategies switch position often, the costs would weigh more on the strategies than on buy-and-hold.
- **Constant interest rate:** Used a constant interest rate for both GBP and RON across the whole 10+ years of data. This is unrealistic, as interest rates fluctuate over time. A more realistic implementation would use historical interest-rate data over the same period. Additionally, real interest rate fluctuations would affect the strategies differently as they spend different amounts of time in each currency. Additionally, the buy-and-hold strategy would also be more complex than simply applying GBP's interest rate cumulatively over the 10+ years, thus providing a more accurate benchmark.
- **Statistical caveats:** Overlapping return windows mean consecutive observations are **not independent**. This means the sample size is smaller than the size of the data frame, making statistical significance harder to establish. Additionally, daily returns are themselves not entirely independent, as market trends influence consecutive days.

Challenges

- **Erroneous Exchange Rate calculation:** I initially took an incorrect approach to applying exchange rates: on each day when the strategy traded, I simply subtracted the Exchange Fee from that day's return, effectively doing $(FX_rate - Fee)$. However, I noticed this isn't symmetrical to the formula used to both apply the FX rate and accumulate interest (used in `Return_InGBP`) - it seemed I was missing the term which describes the relationship between the rate and the FX fee. I thus corrected the formula to $(1 + FX_rate) * (1 - Fee) - 1$, equivalent to $FX_rate - Fee - FX_rate * Fee$ (which corresponds to the form used in `Return_InGBP`).
- **Difficulties working with AI:** I am keen to leverage AI to gain knowledge in fields I do not have exposure to in university. However, I find it difficult to responsibly use AI when I do not fully grasp the matter at hand, because that creates opportunities for hallucinations to go unnoticed. For instance, when starting *Part 2*, I had very little idea how I could model my hypotheses into a simulation - so I used LLMs to explain how this is usually performed. I used the generated code as a starting point - but this led to a significant amount of time spent untangling its logical fallacies, its erroneous underlying assumptions, the aspects I had not clearly specified and which the LLM consequently implemented according to its own interpretation, and mistakes it made. I thus relied more on implementing "by hand" using sources like Stack Overflow, and learning straight from sources like Investopedia going forward to avoid relying on the LLM's limited scope. However, the *dashboard* code is an example where I relied heavily on repeatedly prompting AI to implement my ideas, as I have no knowledge of Streamlit apps and wanted to be efficient given the time I had available. I ensured the code behaves as it should by checking the file after every change and having a high-level understanding of the code to be sure it follows the simulation, and also prompted other LLMs to check for mistakes.

Future extensions

- Implement different entry and exit thresholds or a minimum holding period to minimize the number of trades and consequently decrease the overall effect of numerous exchange fees on the strategies' returns
- Execute trades at the next day's open, calculate return using close/open ratio on that day
- Model bid/ask spreads instead of constant exchange fee
- Add Sharpe Ratio, Information Ratio, time in market to the dashboard
- Add a long/short version, learn how shorting can be modelled
- Research which economic conditions favour each strategy - analyze data over periods of political unrest in Romania (2024 presidential elections, 2026 government crisis) and see if strategies' results hold over these isolated periods of volatility and uncertainty
- Extend to intraday rather than daily trading.

References

- [1] Revolut, *Exchange Fees per Plan* <https://help.revolut.com/help/wealth/exchanging-money/how-much-does-it-cost-to-make-an-exchange/will-i-be-charged-for-exchanging-foreign-currencies/>
- [2] Revolut, *Instant Access Savings* <https://www.revolut.com/savings/savings-calculator/>
- [3] BCR (Romanian Commercial Bank), *Interest Rates on Savings Accounts, Table on bottom of Page 1* https://cdn0.erstegroup.com/content/dam/ro/bcr/www_bcr_ro/Persoane-fizice/Documente-importante/BCR_Dobanzi_pentru_contul_curent_si_produsele_de_economisire_PF.pdf